

Setup Guide — P4A 2026

Complete steps 1–3 before Session 3. Step 4 (VS Code) is optional — only needed if you plan to use it for project work. Step 5 (Ollama) can wait until before Session 5.

1. Download the Course Materials

Download the course folder from Virtual Campus. It contains everything you need: notebooks, documentation, sample datasets, and the environment file.

Place it somewhere easy to find on your machine. The folder should look like this:

```
P4A/
├── environment.yml
├── docs/
├── notebooks/
├── data/
│   └── (sample CSV files for in-class exercises)
```

Keep this folder — you will work in it throughout the course. The project dataset is stored in the course database and accessed directly from the notebooks — no separate download is needed.

2. Opening a Terminal

Several steps below require a terminal — a text-based window where you type commands and press Enter to run them.

macOS — Press **Cmd+Space**, type **Terminal**, press Enter. A window opens with a **\$** prompt.

Linux — Press **Ctrl+Alt+T**, or search for "Terminal" in your application menu.

Windows — After installing Miniconda (next step), use **Anaconda Prompt** — search for it in the Start menu. Do not use the standard Command Prompt or PowerShell for conda commands.

You do not need to understand everything the terminal shows. Type the commands exactly as written and press Enter. If you see an error, copy the message and ask for help.

3. Python Environment

The course uses a shared conda environment that installs all required packages in one step.

Install Miniconda

Official installation guide (macOS, Linux, and Windows):

<https://www.anaconda.com/docs/getting-started/miniconda/install/overview>

macOS

Open Terminal and run:

```
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86\_64.sh  
bash Miniconda3-latest-MacOSX-x86_64.sh
```

Apple Silicon (M1/M2/M3/M4)? Use this installer instead:

```
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh  
bash Miniconda3-latest-MacOSX-arm64.sh
```

Follow the prompts. When asked *"Do you wish the installer to initialize Miniconda3?"* — type **yes**. Then restart your terminal.

Verify:

```
conda --version
```

Linux

Open a terminal and run:

```
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86\_64.sh  
bash Miniconda3-latest-Linux-x86_64.sh
```

Follow the prompts. Type **yes** when asked to initialise. Restart your terminal, then verify:

```
conda --version
```

Windows

1. Download the Miniconda installer: [Miniconda3-latest-Windows-x86_64.exe](#)
2. Run the installer. On the **Advanced Options** screen, tick **"Add Miniconda3 to my PATH environment variable"**
3. Open **Anaconda Prompt** (search for it in the Start menu) — use this for all conda commands

Verify in Anaconda Prompt:

```
conda --version
```

Create the course environment

Open a terminal and navigate to your **P4A/** folder.

- **macOS:** open Terminal, then run:

```
cd path/to/P4A
```

Shortcut: right-click the **P4A** folder in Finder → **New Terminal at Folder**

- **Linux:** open a terminal, then run:

```
cd path/to/P4A
```

- **Windows:** open Anaconda Prompt, then run:

```
cd path\to\P4A
```

Shortcut: hold **Shift** and right-click the **P4A** folder in Explorer → **Open in Terminal**

Confirm you are in the right place — you should see **environment.yml** listed:

```
ls    # macOS / Linux
dir   # Windows
```

Then create the environment:

```
conda env create -f environment.yml
```

This installs Python 3.13 and all course packages. It takes a few minutes the first time.

Activate and launch JupyterLab

```
conda activate p4a2026
jupyter lab
```

JupyterLab opens in your browser. This is the environment used in all class sessions.

4. VS Code (optional — project work only)

JupyterLab is used in all class sessions and the exam. VS Code is optional — useful if you want a richer environment for editing `analysis.py` during the project.

Official installation guide: <https://code.visualstudio.com/docs/setup/setup-overview>

Follow the guide for your operating system. Once installed, come back here to set up the extensions.

Extensions

1. Open the Extensions panel: **Ctrl+Shift+X** on Windows/Linux, **Cmd+Shift+X** on macOS
2. Search for **Python** — install the one published by **Microsoft**
3. Search for **Jupyter** — install the one published by **Microsoft**

Connect VS Code to the course environment

1. Open the command palette: **Ctrl+Shift+P** / **Cmd+Shift+P**
2. Type **Python: Select Interpreter** and press Enter
3. Select **p4a2026** from the list

If **p4a2026** does not appear, make sure you have run `conda env create -f environment.yml` first, then restart VS Code.

5. Ollama — Local LLM (before Session 5)

Ollama runs language models locally on your machine. No API key or internet connection required during use.

Official website: <https://ollama.com>

Install

macOS / Linux — open a terminal and run:

```
curl -fsSL https://ollama.com/install.sh | sh
```

Windows — download and run the installer from <https://ollama.com>

Pull the course model

The course uses **Llama 3.2** — a compact, capable model that runs on most laptops.
Model page: <https://ollama.com/library/llama3.2>

```
ollama pull llama3.2
```

This downloads approximately 2 GB. Do this on a good connection before Session 5.

Verify

```
ollama run llama3.2
```

Type a message and press Enter. If you get a response, you are ready.

Limited disk space? Use `ollama pull llama3.2:1b` instead (~1 GB, slightly less capable).

Explore other models

Ollama has a large library of open models you can download and run locally — different sizes, languages, and specialisations.

Browse the full catalogue: <https://ollama.com/search>

Some models are very large (e.g. Llama 4 starts at 67 GB) and require powerful hardware. Always check the model size before downloading.

Troubleshooting

conda: command not found — Miniconda is not on your PATH. Restart your terminal after installation, or run the Miniconda installer again and select "Add to PATH".

conda env create fails — Make sure you are inside the `P4A/` folder (where `environment.yml` lives) when you run the command. Run `ls` or `dir` to check.

JupyterLab does not open — Confirm the environment is active (`conda activate p4a2026`) before running `jupyter lab`.

ollama: command not found — Restart your terminal after installing Ollama. On macOS, Ollama runs as a background app — check the menu bar icon.

VS Code does not find the p4a2026 environment — Open a terminal inside VS Code and run `conda activate p4a2026`. Then reopen the interpreter selector.

If you are still stuck, bring your laptop to Session 3 and ask during setup time at the start of class.